

Jason Seh

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EDUCATION

B.S. in Computer Science & Engineering, Ohio State University

Expected May 2029

EXPERIENCE

Incoming Software Engineering Intern

Center for Software Innovation x Honda

Columbus, OH

Fall 2026

AI/ML Research Assistant

The Ohio State University | Python, PyTorch, Docker, Kubernetes, GitHub Actions, DINOv2, SAM

Columbus, OH

Apr 2026 - Present

- Developed a distributed AI pipeline on the NRP Nautilus Kubernetes cluster and Tapis HPC platforms for zero-shot image classification, segmentation, and VQA using DINOv2, SAM, and Phi-3-Vision across large-scale image datasets.
- Resolved GPU starvation by redesigning the data ingestion pipeline with multithreaded PyTorch DataLoaders, lazy loading, and model-weight caching, sustaining ~91% GPU utilization during large-scale inference.
- Parallelized stratified cross-validation across 4 GPUs using Python multiprocessing, reducing end-to-end training time by ~75%.
- Engineered GitHub Actions CI/CD pipelines to automatically build and deploy containerized AI microservices using Docker and Kubernetes, enabling reproducible remote batch inference and automated container management.

Technical Program Volunteer – AI/ML

The Ohio Supercomputer Center | Python, PyTorch, DINOv2, YOLO, Slurm, Bash, ONNX, HPC

Columbus, OH

May 2026 - May 2026

- Selected by OSC leadership to develop curriculum, starter codebases, solution repositories, and technical documentation for the “AI in the Field” summer program.
- Converted computer vision models to ONNX and optimized inference pipelines for OSC Ascend and Cardinal HPC clusters, reducing latency by ~66%.
- Trained computer vision models on 10k+ agricultural images and developed an interactive Gradio interface for students to explore classification predictions.

PROJECTS

[OncoPath](#) | *Next.js, FastAPI, PyTorch, XGBoost, SHAP, Timm, Three.js, Claude API*

Feb 2026 - Apr 2026

- Developed 21 XGBoost and PyTorch models using 25k+ MSK-MET patient records to predict organ-specific metastatic risk across 21 anatomical sites, achieving an average 0.72 AUC-ROC.
- Fine-tuned a Phikon Vision Transformer (ViT) on pathology imagery to generate embeddings and fuse histopathology with genomic data for breast cancer risk prediction.
- Built a FastAPI backend for real-time inference and integrated SHAP analysis to explain model predictions using key genomic drivers (e.g., KRAS, SMAD4, TP53).

[AceIt - HackAI](#) | *Next.js, FastAPI, WebRTC, PyTorch, Librosa, Whisper, Gemini, Supabase*

Feb 2026

- Built an AI-powered mock interview platform that used real-time computer vision and audio analysis to evaluate candidate performance across behavioral and technical interviews; placed 7th among 100+ teams.
- Engineered an asynchronous audio pipeline using Whisper, Librosa, and PyTorch, reducing response latency by ~80%
- Built an asynchronous backend with aiohttp and aiortc for low-latency media streaming and concurrent session processing, storing interview analytics in Supabase.

[Buckeye Sense - MakeOhio](#) | *React, Vite, Supabase, Postgis, Deck.gl, MapLibre, ESP32*

Mar 2026

- Built a real-time data pipeline to ingest hashed Wi-Fi telemetry from distributed ESP32 sensors and visualize campus-wide crowd density in 3D for emergency response scenarios.
- Engineered a geospatial processing layer using PostGIS to triangulate and plot device coordinates with sub-second latency.
- Developed asynchronous telemetry-processing pipelines to handle high-frequency sensor data and visualize location clusters for emergency coordination.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, Bash

Frameworks & Libraries: FastAPI, Node.js, Next.js, React, aiohttp

AI/ML: PyTorch, XGBoost, DINOv2, SAM, Phi-3-Vision, ONNX, YOLO, Scikit-Learn, Computer Vision, Multimodal AI

Infrastructure & Tools: Docker, Grafana, Slurm, Git, GitHub Actions, Linux, Vercel, Cloudflare R2

Database: PostgreSQL, PostGIS, Supabase, MongoDB